

Uncovering Spatial Patterns in Singapore's Flat Resale Transactions

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Abstract—This report investigates spatial patterns in Singapore's Housing & Development Board(HDB) resale transactions through a dual approach of spatial statistical analysis and machine learning. Spatial statistical methods, including Moran's I, LISA, variograms, KDE, Geary's C, Getis-Ord G* and Temporal Hotspot Analysis, uncover global and local clustering, spatial ranges of influence, and price density gradients. The machine learning component employs feature engineering, Geographically Weighted Regression, HDBSCAN clustering, and Random Forest modeling. The analysis integrates spatial autocorrelation, heterogeneity, and predictive modeling. The report details dataset preprocessing, methodological justifications, visualizations, and challenges, providing insights into the spatial and socio-economic dynamics of Singapore's resale market, with a focus on the critical role of proximity and neighborhood price effects.

I. DATASET DESCRIPTION

This dataset [1] is retrieved from Singapore's national data repository on 30 Apr 2025. It is an accumulation of information relating to the sale of Singapore's public housing apartments colloquially referred to as flats.

A. Dataset Structure

Each record represents a single flat resale transaction, with attributes categorized as:

- **Transaction Details:** Identifiers and temporal information.
- **Location Information:** Spatial coordinates and administrative boundaries.
- **Flat Characteristics:** Physical and lease-related properties.
- **Proximity Features:** Distances to key amenities.

B. Key Attributes

1) Transaction Details:

- `transaction_id`: Unique identifier for each transaction.
- `month` (e.g., 2017-01-01): Date of the resale transaction.
- `resale_price` (e.g., S\$288,500): Sale price of the flat.
- `price_per_sqft` (e.g., 394.16): Price per square foot, derived from resale price and floor area.

2) Location Information:

- `town` (e.g., Ang Mo Kio): Administrative town of the flat.
- `blk_no`, `road_name`, `building`, `postal` (e.g., 534, Ang Mo Kio Avenue 10, Cheng San View, 560534): Address details.
- `planning_area_ura` (e.g., Ang Mo Kio): Urban Redevelopment Authority planning area.
- `region_ura` (e.g., North-East Region): Broader regional classification.
- `x`, `y` (e.g., 30320.30, 39561.93): Spatial coordinates in a projected system (SVY21/EPSSG:3414).
- `latitude`, `longitude` (e.g., 1.374058, 103.854168): Geographic coordinates for geospatial analysis.

3) Flat Characteristics:

- `storey_range` (e.g., 01 to 03): Floor range of the flat.
- `flat_type` (e.g., 3 Room): Number of rooms or flat category.
- `flat_model` (e.g., New Generation): Design model of the flat.
- `lease_commence_date` (e.g., 1980-02-01): Start date of the 99-year lease.
- `remaining_lease_years`, `remaining_lease_months` (e.g., 62 years, 1 month): Remaining lease duration.
- `floor_area_sqm`, `floor_area_sqft` (e.g., 68.0 m², 731.95 ft²): Floor area of the flat.

4) Proximity Features:

- `closest_mrt_station` (e.g., Ang Mo Kio): Nearest Mass Rapid Transit (MRT) station.
- `distance_to_mrt_meters` (e.g., 686.53): Distance to the closest MRT station.
- `transport_type` (e.g., MRT): Type of transport hub.
- `line_color` (e.g., Red): MRT line color for the closest station.
- `distance_to_cbd` (e.g., 9948.87 meters): Distance to the Central Business District.
- `closest_pri_school` (e.g., Jing Shan Primary School): Nearest primary school.

- `distance_to_pri_school_meters` (e.g., 303.45): Distance to the closest primary school.

II. EXPLORATORY SPATIAL STATISTICAL ANALYSIS

A. Understanding the Data and Preprocessing

The Singapore flat resale transactions dataset, with 203,196 records and 30 attributes, supports exploratory spatial analytics by capturing spatial and non-spatial features of public housing sales. This subsection outlines its structure, key attributes, preprocessing steps, and implementation details.

Dataset Structure: The dataset has 203,196 rows and 30 columns, with numerical attributes (e.g., `resale_price`, `x`, `y`) as float64/int64 and categorical attributes (e.g., `town`, `flat_type`).

Key Attributes:

- **Spatial:** `x`, `y` (SVY21/EPSC:3414), `latitude` (1.2704–1.4571), `longitude` (103.6852–103.9878), `distance_to_mrt_meters`, `distance_to_cbd`, `distance_to_pri_school_meters`.
- **Non-Spatial:** `resale_price`, `price_per_sqft`, `town` (26 unique), `flat_type`, `floor_area_sqm`, `planning_area_ura`, `closest_mrt_station`, etc.

Summary Statistics:

- Correlation analysis (Figure 1): `distance_to_cbd` shows a strong positive correlation with `y` (0.77), indicating northern areas (higher `y`) are farther from the CBD. `price_per_sqft` has a moderate negative correlation with `distance_to_cbd` (-0.45) and `distance_to_mrt_meters` (-0.16), suggesting higher prices near the CBD and MRT stations, supporting the hypothesis of spatial clustering. `distance_to_pri_school_meters` shows a slight positive correlation with `price_per_sqft` and `resale_price`, hinting at a lower contribution to prices.

Preprocessing Steps:

- Filtered to keep only the latest listing per flat, reducing the dataset to 69,582 rows. This ensures a unique spatial snapshot for each flat, as multiple listings over time could skew spatial patterns (e.g., repeated `x`, `y` coordinates with outdated prices), which is critical for accurate spatial analysis.
- No missing values or duplicates detected.
- Verified `x`, `y` to align with SVY21 (EPSG:3414) and `latitude`, `longitude` ranges.

Implementation Details: The dataset was loaded using `pandas.read_csv`. Duplicates were checked with `df.duplicated` (none found), and summary statistics were generated with `df.describe`. Spatial attributes (`x`, `y`, `latitude`, `longitude`, etc.) and non-spatial attributes were identified. Missing values were assessed with `df.isnull().sum` (none found), and coordinate ranges were verified. A correlation matrix for `x`, `y`,



Fig. 1. Correlation matrix of spatial and price features, highlighting relationships between `distance_to_cbd`, `distance_to_mrt_meters`, `price_per_sqft`, and spatial coordinates.

`distance_to_mrt_meters`, `distance_to_cbd`, `distance_to_pri_school_meters`, `resale_price`, and `price_per_sqft` was computed with `df.corr` and visualized using `seaborn.heatmap` with a `coolwarm` colormap.

B. Spatial Pattern Discovery

1) **Global Spatial Autocorrelation:** Global spatial autocorrelation was assessed to test the hypothesis that high resale prices are spatially clustered, potentially due to proximity to the CBD or MRT stations. Moran's I and Geary's C statistics were computed for `resale_price` and `price_per_sqft`.

Rationale for Methods: Moran's I was selected as the main measure to estimate global spatial autocorrelation because it is powerful in identifying clustering in continuous data such as prices. Moran's I quantifies the extent of similarity between spatially neighboring observations, consistent with the clustering hypothesis around important amenities. Geary's C was employed as a supplementary measure to check against local dissimilarity, considering squared differences in neighboring values, which gives another view of spatial dependence. Both measures are responsive to the spatial weights' selection, and therefore they are applicable for testing the hypothesis.

Implementation Details: The analysis was performed utilizing Python using `geopandas`, `libpysal`, and `esda` libraries. The dataset was loaded into a `GeoDataFrame` utilizing `x` and `y` coordinates of Singapore's SVY21 projection (EPSG:3414) to maintain the proper spatial relations. A k-nearest neighbors (`k=40`) spatial weights matrix was produced using `KNN.from_dataframe`, with `k=40` being an optimal choice between the local spatial relations and the computational efficiency for the large database (203,196 records). The

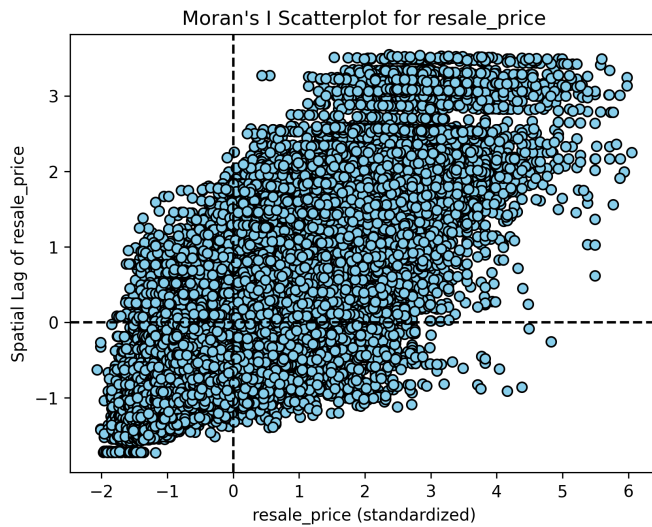


Fig. 2. Moran's I scatter plot for `resale_price`, showing strong positive spatial autocorrelation ($I = 0.6517$, $p = 0.0010$).



Fig. 3. Moran's I scatter plot for `price_per_sqft`, indicating very strong positive spatial autocorrelation ($I = 0.7370$, $p = 0.0010$).

weights were row-standardized to standardize the neighbor effect for making influences comparable between regions with different neighbor numbers. Moran's I and Geary's C were calculated for `resale_price` and `price_per_sqft`.

Moran's I Results:

- `resale_price`: $I = 0.6517$, $p = 0.0010$.
- `price_per_sqft`: $I = 0.7370$, $p = 0.0010$.

The Moran's I scatter plots (Figures 2 and 3) plot the standardized variable against its spatial lag and feature a positive slope with the majority of points in the High-High and Low-Low quadrants, reflecting strong spatial clustering. The greater I value for `price_per_sqft` (0.7370) versus `resale_price` (0.6517) reflects stronger clustering, possibly due to the former normalizing for flat size, thereby diminishing variability. Both p-values (0.0010) also affirm statistical significance, corroborating the hypothesis of spatial clustering closer to the CBD and MRT stations.

Visualization Details: These scatter plots were plotted with the help of `matplotlib`, using standardized values of `resale_price` and `price_per_sqft` on the x-axis and their spatial lags on the y-axis. Points were plotted with sky-blue fill and black edges for visibility and had vertical and horizontal dashed lines at zero to separate quadrants (High-High, Low-Low, High-Low, Low-High). This map emphasizes the clustering pattern, where the high concentration in the High-High quadrant signifies that high-priced flats are close to other high-priced flats, in line with the hypothesis.

Geary's C Results:

- `resale_price`: $C = 0.3468$, $p = 0.0010$.
- `price_per_sqft`: $C = 0.2640$, $p = 0.0010$.

Geary's C values less than 1 (0.3468 for `resale_price`, 0.2640 for `price_per_sqft`) reflect positive spatial autocorrelation, in agreement with Moran's I results. The smaller C value for `price_per_sqft` supports its more intense

clustering, as smaller squared differences between neighbors reflect higher similarity. The p-values (0.0010) affirm significance.

Conclusions: These findings validate the hypothesis of spatial clustering, likely driven by proximity to amenities like the CBD and MRT stations, and set the stage for local autocorrelation analysis (e.g., LISA) to identify specific clusters and hotspots.

2) *Local Indicators of Spatial Association (LISA)*: Local Indicators of Spatial Association (LISA) analysis was conducted to identify specific clusters and hotspots of `resale_price` and `price_per_sqft`, building on the global autocorrelation findings. This analysis helps pinpoint areas of high and low price clustering across Singapore.

Rationale for Methods: Local Moran's I was chosen for the identification of local spatial autocorrelation, as it detects both clusters (High-High, Low-Low) and spatial outliers (High-Low, Low-High) at every site. It breaks down the global Moran's I into contributions based on location, which shows spatial heterogeneity in price patterns. Getis-Ord G^* was employed as a second measure to define hotspots (high-value clusters) and coldspots (low-value clusters) through Z-scores, presenting a different view of spatial clustering. Both techniques are sensitive to the spatial weights' choice, thus appropriate for finding localized patterns within the dataset. A threshold of $p < 0.05$ was used to emphasize statistically significant clusters.

Implementation Details: The analysis used the same GeoDataFrame and k-nearest neighbors ($k=40$) spatial weights matrix (row-standardized) as the global analysis, ensuring consistency. Local Moran's I and Getis-Ord G^* were computed for `resale_price` and `price_per_sqft` using `esda.Moran_Local` and `esda.G_Local`. No islands (zero-neighbor points) were reported, confirming the weights matrix's connectivity. Significance was assessed with p-values

from 999 permutations, and significant clusters $p < 0.05$ were mapped. The `splot.esda.lisa_cluster` function was used to generate cluster maps, while custom plotting functions mapped significance and G^* Z-scores.

Results and Interpretation: The LISA cluster maps (Figures 4 and 5) show significant High-High (HH) clusters for both variables in central right diagonal areas like Downtown Core, Punggol, and Queenstown, indicating concentrations of high-priced flats. Low-Low (LL) clusters are observed in peripheral areas such as Jurong West, Woodlands, and Choa Chu Kang, reflecting areas with lower prices. The significance maps (Figures 6 and 7) indicate that most points are statistically significant $p < 0.05$, with `price_per_sqft` showing more pronounced HH clusters, consistent with its stronger global autocorrelation ($I = 0.7370$).

The Getis-Ord G^* Z-score maps (Figures 8 and 9) identify hotspots in central right tilted diagonal areas like Downtown Core, Punggol, and Queenstown with high Z-scores (0.57 to 3.55 for `resale_price`, 0.52 to 4.12 for `price_per_sqft`). Coldspots are observed in areas like Jurong West, Woodlands, and Sembawang, with Z-scores ranging from -1.73 to -0.30 for `resale_price` and -1.36 to -0.10 for `price_per_sqft`, aligning with the LISA LL clusters.

Visualization Details: Three types of maps were generated for each variable: noitemsep

- **Cluster Maps:** Use `splot.esda.lisa_cluster` to plot HH (red), LL (blue), HL (orange), and LH (light blue) clusters at $p < 0.05$, with non-significant points in gray. Town labels were added for geographic context, and the color scheme ensures clear differentiation of cluster types.
- **Significance Maps:** Plot binary significance $p < 0.05$ using a `coolwarm` cmap (red for significant, blue for non-significant), highlighting the prevalence of significant clusters.
- **Getis-Ord G^* Z-Score Maps:** Use a `coolwarm` cmap with a quantile scheme to map Z-scores, ranging from coldspots (blue) to hotspots (red), providing a gradient of clustering intensity.

All maps were saved at 300 DPI with a 10x8 inch size for clarity, using `matplotlib` with consistent town labels to facilitate interpretation across Singapore's geography.

Conclusions: These results highlight localized clustering of high-priced flats in areas near the right tilted center diagonal and low-priced flats in peripheral areas. The findings set the stage for further analysis of spatial heterogeneity and stationarity.

3) **Spatial Heterogeneity and Stationarity:** This analysis examines spatial heterogeneity and stationarity in `resale_price` and `price_per_sqft` using variograms and spatial correlograms. These methods assess how spatial dependence varies with distance and whether the pricing patterns exhibit stationarity across Singapore, building on the global and local autocorrelation findings.

Rationale for Methods: Variograms were employed to quantify spatial heterogeneity by measuring the semivariance

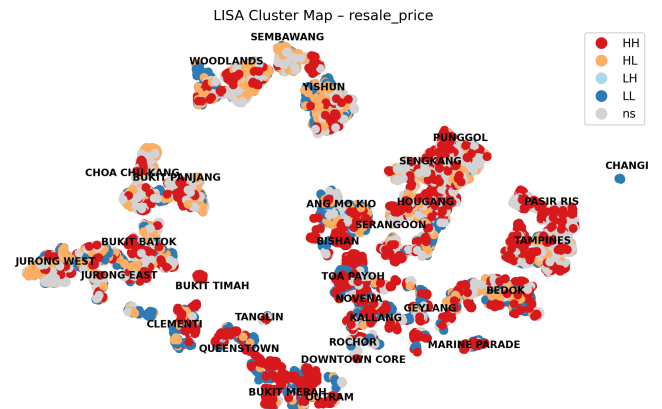


Fig. 4. LISA cluster map for `resale_price`, showing clusters.

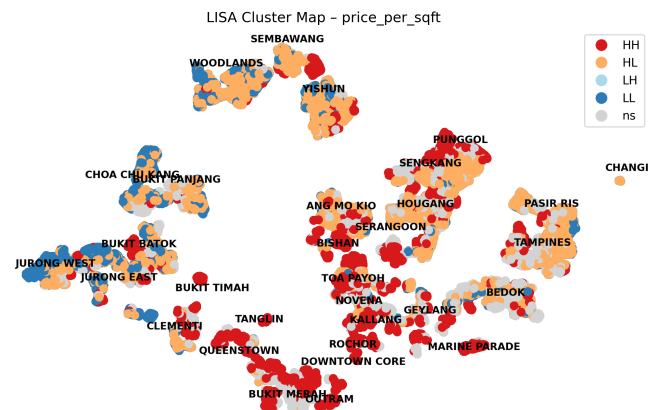


Fig. 5. LISA cluster map for `price_per_sqft`.

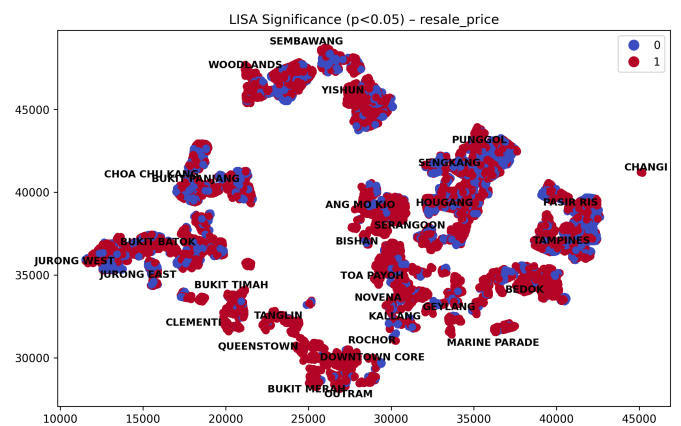


Fig. 6. LISA significance map ($p < 0.05$) for `resale_price`, showing most points as statistically significant.

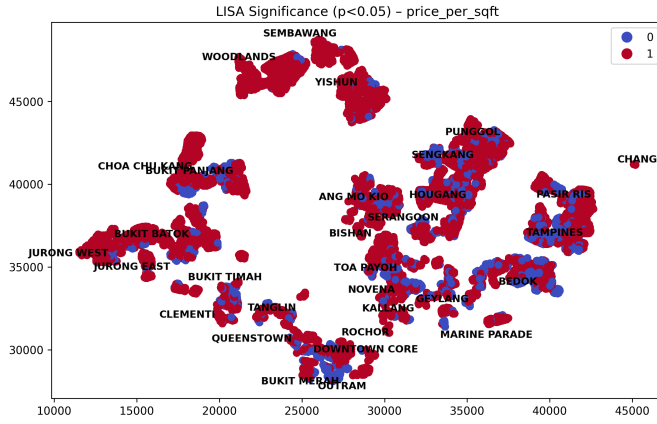


Fig. 7. LISA significance map ($p < 0.05$) for `price_per_sqft`, with widespread significance and stronger clustering patterns.

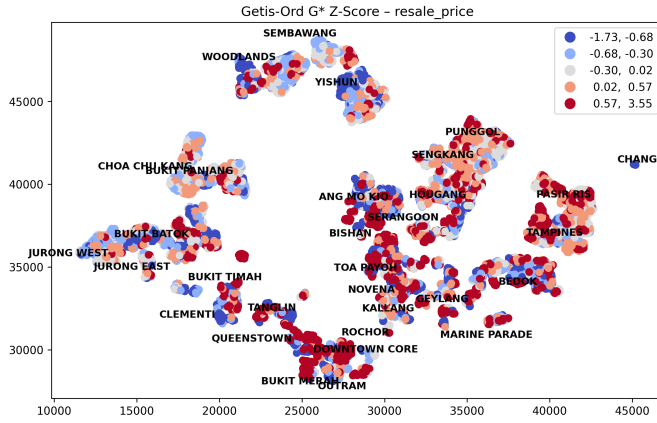


Fig. 8. Getis-Ord G^* Z-score map for `resale_price`, identifying hotspots and coldspots.

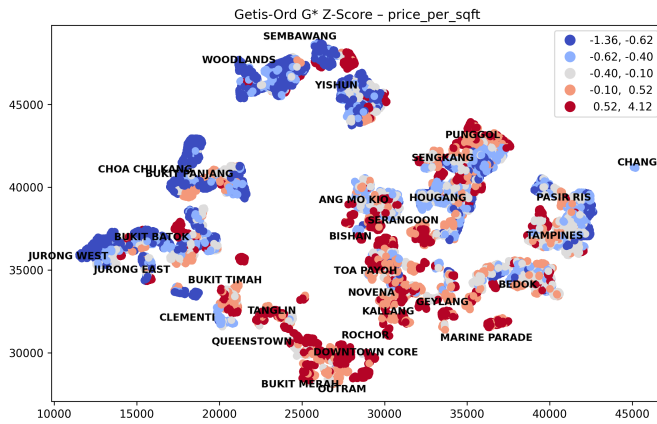


Fig. 9. Getis-Ord G^* Z-score map for `price_per_sqft`.

of price differences as a function of distance (lag), aligning with Section 4. This method captures how price variability changes spatially, indicating whether the data exhibits stationarity (constant mean and variance over space). Spatial correlograms, based on Moran's I across distance bands, were used to evaluate the decay of spatial autocorrelation with distance, providing insight into the scale of spatial dependence. Both methods are robust for detecting non-stationarity and spatial structure in continuous data like prices, as discussed in Section 4.

Implementation Details: The analysis used the same GeoDataFrame as previous sections, with coordinates in Singapore's SVY21 projection (EPSG:3414). For computational efficiency, a random subsample of 50,000 points was selected for each variable using `numpy.random.choice` with a fixed seed (42) for reproducibility. Variograms were computed using `skgstat.Variogram` with 20 lags to capture fine-scale spatial variation, and semivariance was plotted against distance. Spatial correlograms were generated using `libpysal.weights.DistanceBand` and `esda.Moran`, calculating Moran's I for distance bands from 500 m to 20,000 m in 500 m increments. Binary weights were used to focus on presence/absence of neighbors within each distance band.

Results and Interpretation: The variograms for `resale_price` (Figure 10) and `price_per_sqft` (Figure 11) show a rapid increase in semivariance at short distances, with a major change in slope between 3,000 m and 4,000 m. Beyond this range, the semivariance begins to stabilize, reaching approximately 3 m^2 for `price_per_sqft` and 10,000,000 for `resale_price` by 10,000 m. This indicates spatial heterogeneity, as price variability increases significantly up to 3,000–4,000 m, after which the rate of change slows, suggesting a spatial range of influence around this distance. The higher semivariance for `resale_price` reflects its larger absolute values compared to `price_per_sqft`, but both suggest non-stationarity, as the variance is not constant across space.

The spatial correlograms (Figures 12 and 13) show Moran's I starting at approximately 0.20 for `resale_price` and 0.40 for `price_per_sqft` at short distances (500 m). A notable change in slope occurs between 3,000 m and 4,000 m, where the rate of decay in autocorrelation accelerates, dropping more rapidly toward 0 by 20,000 m. This indicates that spatial autocorrelation is strongest at shorter distances and diminishes significantly beyond 3,000–4,000 m, with `price_per_sqft` exhibiting stronger spatial dependence at all scales, consistent with its higher global Moran's I (0.7370 vs. 0.6517 for `resale_price`). The correlograms highlight a spatial structure where price similarity decreases markedly after this distance range, further supporting the presence of spatial heterogeneity and non-stationarity.

Visualization Details: Two types of plots were generated for each variable: `noitemsep`

- **Variograms:** Plot semivariance against distance using `skgstat.Variogram.plot`, with 20 lags for

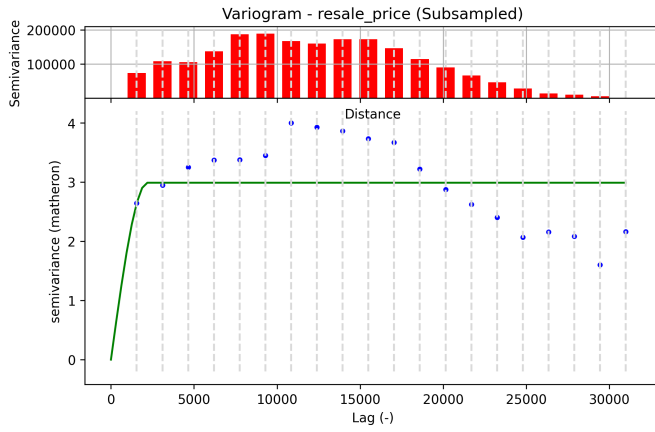


Fig. 10. Variogram for `resale_price` (subsampled), showing a major change in slope between 3,000–4,000 m, indicating spatial heterogeneity.

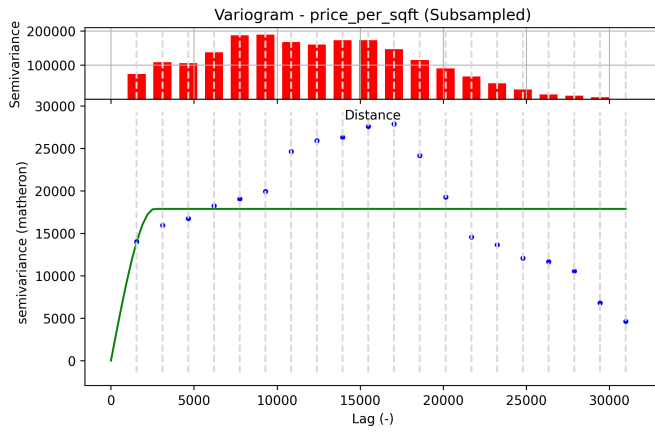


Fig. 11. Variogram for `price_per_sqft` (subsampled), with a significant slope change at 3,000–4,000 m, suggesting a spatial range of influence.

granularity. The y-axis shows semivariance (m^2 for `price_per_sqft`, unitless for `resale_price`), and the x-axis shows distance (m). A green fitted curve highlights the trend, with red bars indicating bin counts for each lag.

- *Spatial Correlograms*: Plot Moran's I against distance bands (500 m to 20,000 m) using `matplotlib`. Points are marked with sky-blue circles connected by a line, with the x-axis in meters and the y-axis showing Moran's I values, providing a clear view of autocorrelation decay.

All plots were saved at 300 DPI with consistent sizing for clarity, using `matplotlib` with grid lines to aid interpretation.

Conclusions: These findings confirm spatial heterogeneity and non-stationarity in the dataset, with a significant change in spatial dependence occurring at 3,000–4,000 m. This sets the stage for further exploration of spatial patterns using kernel density estimation and join count analysis.

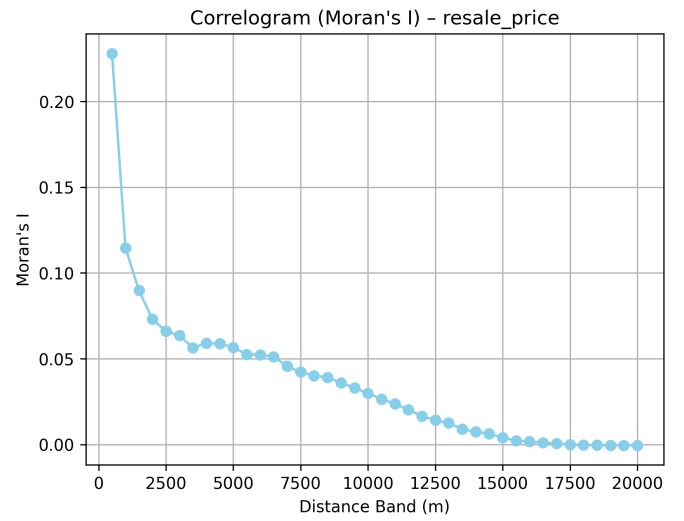


Fig. 12. Spatial correlogram (Moran's I) for `resale_price`, showing a notable decay in autocorrelation between 3,000–4,000 m.

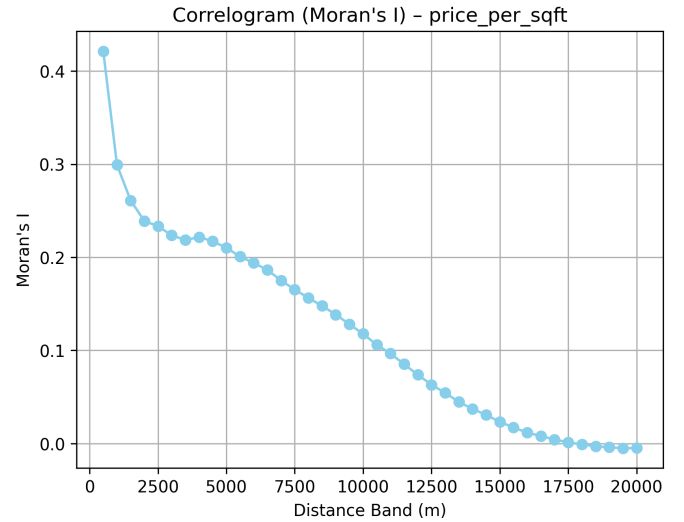


Fig. 13. Spatial correlogram (Moran's I) for `price_per_sqft`, with a marked change in autocorrelation decay at 3,000–4,000 m.

C. Kernel Density Estimation (KDE)

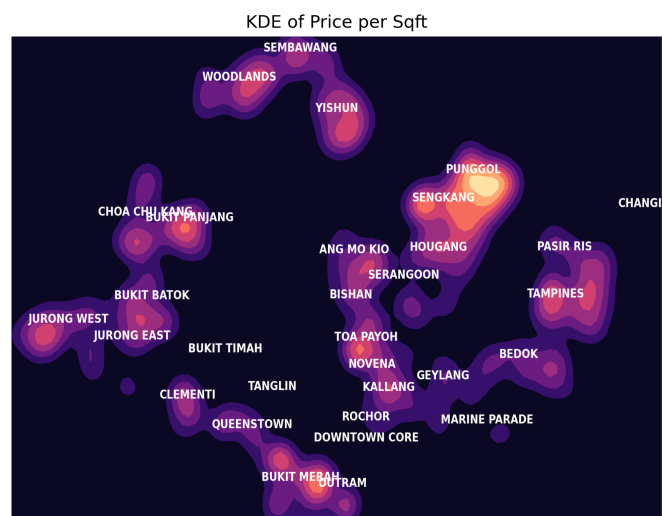
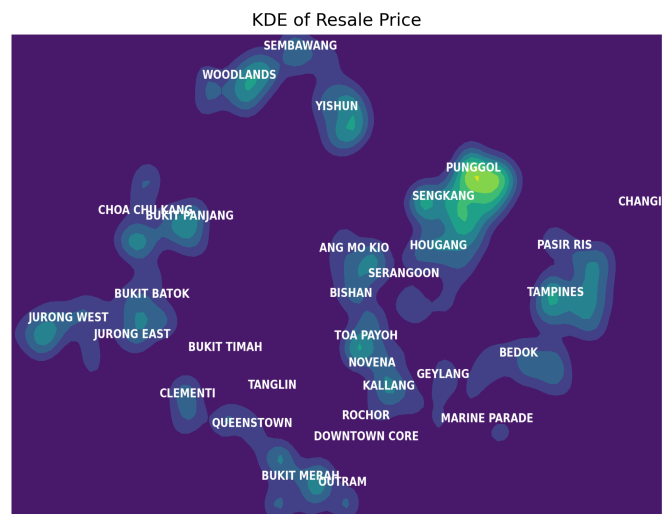
This section applies Kernel Density Estimation (KDE) to explore spatial patterns in `resale_price`, `price_per_sqft`, and proximity to key amenities (MRT, primary schools, and CBD) across Singapore. KDE is used to visualize the intensity of these variables, highlighting areas of high concentration and their spatial relationships, building on the heterogeneity and clustering analyses from previous sections.

Rationale for Methods: KDE was chosen to estimate the continuous spatial density of `resale_price`, `price_per_sqft`, and inverse distances to MRT, primary schools, and CBD, as it provides a smooth representation of point intensity. This method is effective for identifying spatial

concentrations, such as areas with high prices or proximity to amenities, and complements the LISA and variogram analyses by offering a non-parametric approach to spatial pattern discovery. A Gaussian kernel with a 2000 m bandwidth was selected to balance local detail and overall trends, aligning with the spatial range of influence (3,000–4,000 m).

Results and Interpretation: The KDE map for `resale_price` (Figure 14) shows high-density regions in central areas like Senggang, Punggol, and Bukit Merah, indicating concentrations of higher resale prices. The KDE for `price_per_sqft` (Figure 15) similarly highlights high-density clusters in Senggang, Punggol, and Bukit Merah, with `price_per_sqft` showing more pronounced concentration in these regions, consistent with its stronger spatial autocorrelation (Moran's $I = 0.7370$ vs. 0.6517 for `resale_price`).

Visualization Details: Five KDE maps were generated:



closest to schools.

All maps were plotted using `matplotlib.contourf` on a 100x100 grid, with town labels added for geographic context. Maps were saved at 300 DPI with an 8x6 inch size, with axes turned off to focus on spatial patterns.

D. Temporal Hotspot Analysis

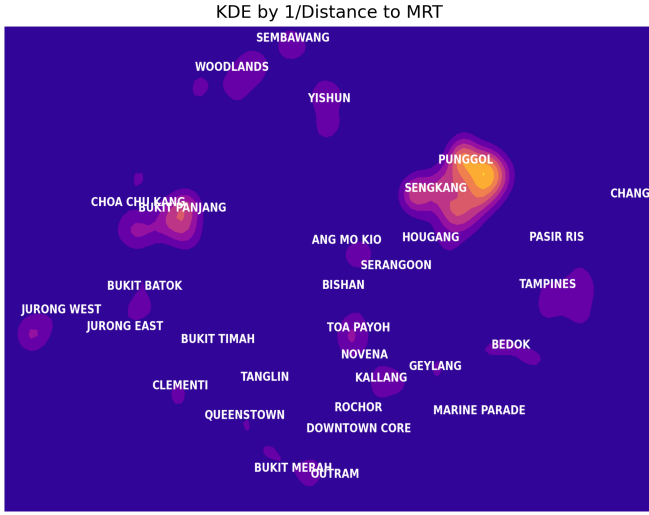


Fig. 16. KDE of 1/distance to MRT.

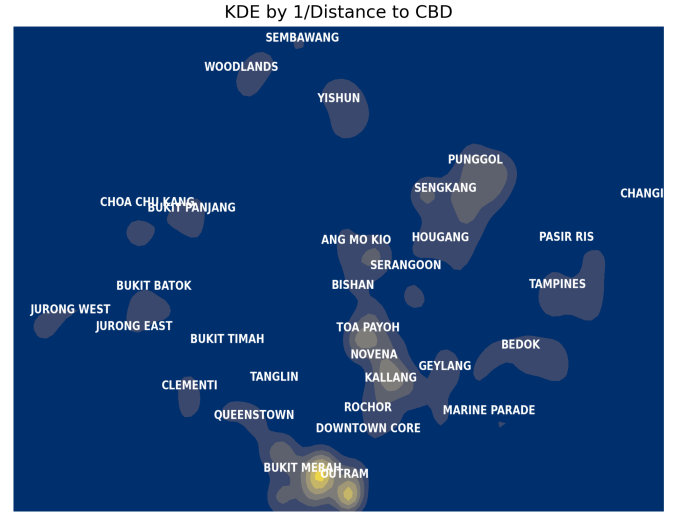


Fig. 18. KDE of 1/distance to CBD.

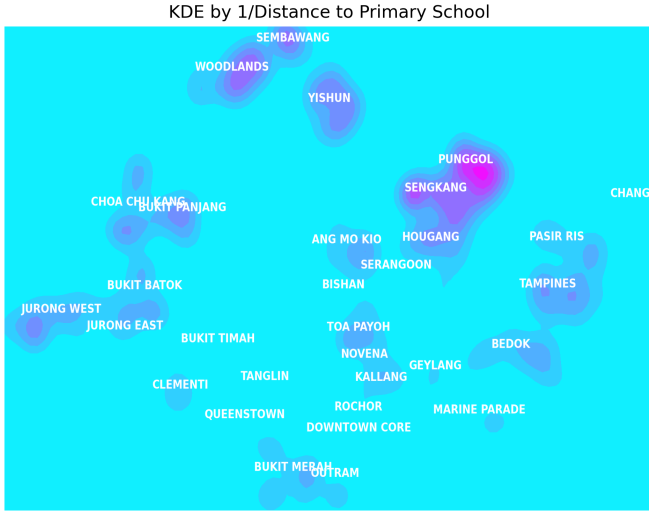


Fig. 17. KDE of 1/distance to primary school.

for its ability to provide smooth density estimates, highlighting hotspots without predefined boundaries. We analyze `price_per_sqft` using the oldest and latest resale transactions to assess temporal shifts. A Gaussian kernel with a 2000 m bandwidth balances local and broader trends.

Implementation Details The analysis used a GeoDataFrame in Singapore’s SVY21 projection (EPSG:3414). A 100×100 grid was created over the data’s bounding box using `numpy.linspace` and `meshgrid`. The `sklearn.neighbors.KernelDensity` class with a Gaussian kernel and 2000 m bandwidth was applied to `price_per_sqft`. Density estimates were exponentiated for visualization. The oldest and latest transactions were used for each KDE.

Results and Interpretation KDE results show minimal change in hotspots over time, consistent with previous sec-

tions. High-density areas for `price_per_sqft` remain the same for both the oldest and latest transactions. Proximity to CBD, MRT and schools continues to drive higher price densities, with no significant temporal shift. The stability suggests persistent demand in these areas, likely due to established infrastructure and amenities.

Visualization Details:

- *KDE of `price_per_sqft`*: (Figure 15): Uses the magma colormap (purple to yellow), highlighting areas of high price per square foot in yellow.
- *KDE of Oldest `price_per_sqft`*: (Figure 19): Uses the magma colormap (purple to yellow), highlighting areas of high price per square foot in yellow.

All maps were plotted using `matplotlib.contourf` on a 100×100 grid, with town labels added for geographic context. Maps were saved at 300 DPI with an 8x6 inch size, with axes turned off to focus on spatial patterns.

Conclusions: Hotspots for housing prices in Singapore remain stable over time. Proximity to CBD, MRT and schools remains a key factor. This stability aligns with previous findings, suggesting sustained demand in these areas.

E. Conclusion and Final Remarks

This exploratory spatial statistical analysis uncovered significant patterns in resale transactions across Singapore, focusing on `resale_price`, `price_per_sqft`, and proximity to MRT, schools, and CBD. Variograms identified a spatial range of influence of 3,000–5,000 meters. LISA analysis (Moran’s I: 0.6517 for `resale_price`, 0.7370 for `price_per_sqft`) revealed clustering of high-priced flats, with lower clusters in peripherals. KDE confirmed these patterns, showing high-density regions for `resale_price` and `price_per_sqft` in Sengkang, Punggol etc, driven by proximity to MRT, CBD and schools.

These findings highlight the role of transport accessibility in driving price premiums, aligning with Singapore’s transit-

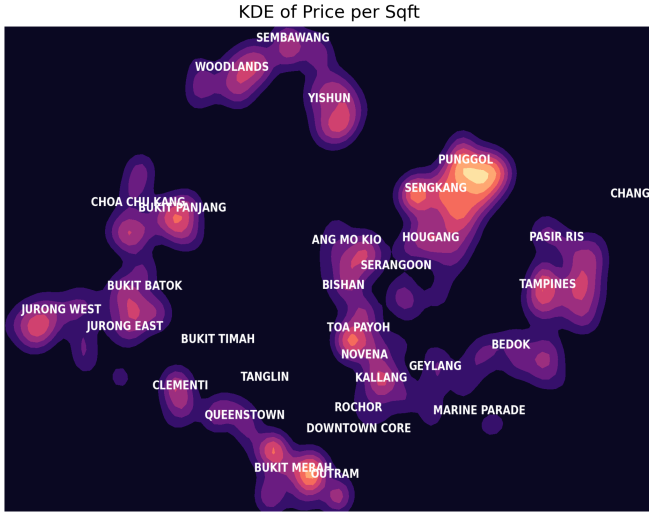


Fig. 19. KDE of Price per Sqft (Oldest Transactions).

oriented development, while the even distribution of schools suggests equitable planning. The 3,000–5,000 meter range offers a practical scale for future urban planning to ensure amenity access and price stability.

The minimal temporal effect on hotspots highlights the stability of housing demand in regions, likely due to consistent infrastructure and accessibility advantages, reinforcing the enduring influence of spatial factors over temporal dynamics in Singapore’s resale market.

This sets the stage for SPATIAL MACHINE LEARNING, where regression models can quantify the impact of spatial factors on `resale_price`, with MRT and CBD proximity likely as key predictors. Future work could explore finer temporal granularity to detect subtle shifts.

III. SPATIAL MACHINE LEARNING

A. Feature Engineering

Rationale for Methods: Feature engineering was designed to capture spatial relationships and key predictors of `price_per_sqft` while minimizing temporal biases. We filtered the dataset to include only the latest listing per flat, ensuring a unique spatial snapshot. Distance-based features (e.g., to MRT, CBD, schools) were standardized to normalize their scales for regression modeling. Spatial lag features were introduced using a k-nearest neighbors ($k=40$) weights matrix to account for neighborhood effects, as spatial autocorrelation was evident from the LISA analysis (Moran’s I : 0.7370 for `price_per_sqft`). Categorical variables like planning area and flat type were encoded to incorporate regional and property-type variations.

Implementation: The dataset, initially containing HDB resale transactions, was filtered to retain only the latest listing per flat, reducing it to 69,582 rows. A `GeoDataFrame` was created with SVY21 projection (EPSG:3414) using x and y coordinates. Distance features (`distance_to_mrt_meters`, `distance_to_cbd`,

`distance_to_pri_school_meters`) were standardized using `StandardScaler`. A spatial lag for `price_per_sqft` was computed using a k-nearest neighbors weights matrix ($k=40$) via `libpysal`. Categorical features (`planning_area_ura`, `flat_type`) were label-encoded. The final feature set included distance-based features, the `price_per_sqft` spatial lag, encoded categoricals, `floor_area_sqm`, with `price_per_sqft` as the target. Rows with missing values were dropped, and the data was split into training (80%) and testing (20%) sets.

Results and Interpretation: The filtered dataset of 69,582 rows ensured a spatially unique representation of flats, avoiding bias from repeated listings. Standardization of distance features enabled fair comparisons across variables with different scales. The spatial lag feature `price_per_sqft_lag` captured neighborhood effects, reflecting the influence of nearby flats’ prices, aligning with the observed spatial autocorrelation. Encoded categorical features introduced regional and flat-type variations, while `floor_area_sqm` added structural context without focusing on transaction timing. The train-test split resulted in 55,665 training samples and 13,917 testing samples, providing a robust dataset for modeling.

Conclusions: Feature engineering successfully prepared the dataset for spatial machine learning by focusing on spatial predictors and minimizing temporal influences, consistent with the minimal temporal effect observed. The inclusion of the `price_per_sqft` spatial lag and standardized distance features ensures that regression models can effectively capture the spatial dynamics driving `price_per_sqft`, such as proximity to MRT and CBD. This sets a strong foundation for predictive modeling.

B. Regression Modelling

1) *OLS Regression:* Ordinary Least Squares (OLS) regression is a fundamental statistical method used to model the linear relationship between a dependent variable and one or more independent variables. It minimizes the sum of squared residuals to estimate the best-fitting line, making it widely used for its simplicity and interpretability in predictive modeling. In spatial contexts, OLS can serve as a baseline to evaluate the influence of spatial features, though it assumes independence of observations, which may be violated due to spatial autocorrelation.

Rationale for Methods: Ordinary Least Squares (OLS) regression was chosen as the baseline model to quantify the impact of spatial features on `price_per_sqft`, given its simplicity and interpretability. The model leverages the engineered features, including distance-based variables, spatial lags, and categorical encodings, to capture spatial dynamics identified in the exploratory analysis. Residual analysis, including spatial autocorrelation via Moran’s I , was conducted to assess model assumptions and detect unmodeled spatial effects, aligning with the observed spatial range of influence (3,000–5,000 meters) and clustering patterns.

Implementation: An OLS regression model was fitted using `sklearn.linear_model.LinearRegression` on

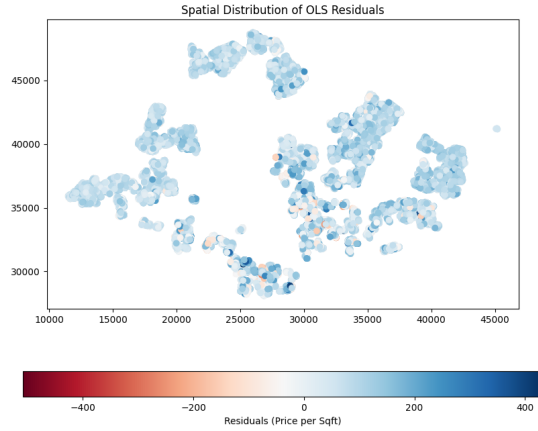


Fig. 20. Spatial Distribution of OLS Residuals.

the training set (55,665 samples) from the feature engineering step. Predictions were made on both training and test sets (13,917 samples). Performance metrics, including R^2 , adjusted R^2 , mean squared error (MSE), and root mean squared error (RMSE), were computed for both sets. Residuals were calculated for the full dataset to analyze model fit. Spatial autocorrelation of residuals was assessed using Moran's I with a k-nearest neighbors ($k=40$) weights matrix via `libpysal`. Two visualizations were generated: a scatter plot of residuals versus predicted values and a spatial map of residuals.

Results and Interpretation: The OLS model achieved an R^2 of 0.7092 (adjusted R^2 : 0.7091) on the training set and 0.7076 (adjusted R^2 : 0.7074) on the test set, indicating consistent performance and explaining approximately 71% of the variance in `price_per_sqft`. The RMSE was 81.4875 on the training set and 82.6628 on the test set, suggesting moderate prediction errors relative to the target's scale (300–1100). Residual analysis showed no significant spatial autocorrelation (Moran's I: 0.0011, p-value: 0.1010), suggesting that the model adequately captures spatial dependencies.

Visualization Details: Two plots were generated to evaluate the model. Figure 21 shows a scatter plot of residuals versus predicted `price_per_sqft`, with a red dashed line at zero. The plot reveals a relatively symmetric distribution around zero, with residuals ranging from -400 to 400, indicating no systematic bias but some heteroscedasticity at higher predictions. Figure 20 displays the spatial distribution of residuals using a red-blue colormap, showing a more random pattern across Singapore compared to earlier analyses, consistent with the insignificant Moran's I.

Conclusions: The OLS model provides a solid baseline for predicting `price_per_sqft`, capturing 71% of the variance with consistent performance across training and test sets. The lack of significant spatial autocorrelation in residuals indicates that the model effectively accounts for spatial dependencies,

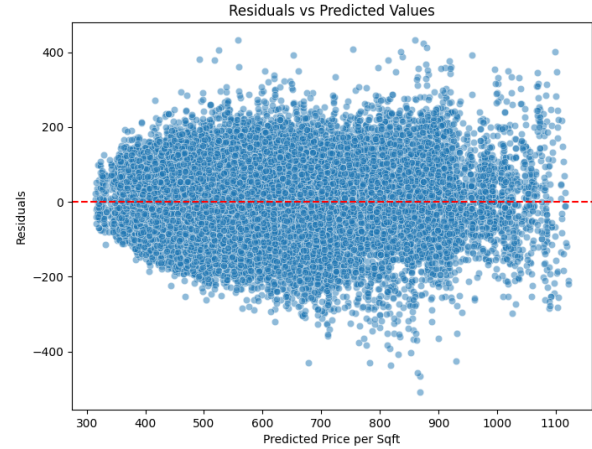


Fig. 21. Residuals versus Predicted Values for OLS Regression.

likely due to the inclusion of the `price_per_sqft` spatial lag feature. The visualizations confirm a random residual pattern, aligning with the exploratory findings of stable clustering in areas like Sengkang and Punggol. While the model performs well, exploring more complex models like spatial lag or error models could further improve prediction accuracy by addressing potential nonlinear relationships.

2) Geographically Weighted Regression (GWR): Geographically Weighted Regression (GWR) extends traditional regression by allowing model parameters to vary spatially, capturing local relationships between variables. Unlike OLS, which assumes spatially constant relationships, GWR uses a spatially weighted kernel to estimate local coefficients, making it suitable for modeling spatially heterogeneous processes like housing prices in Singapore.

Rationale for Methods: GWR was employed to address spatial heterogeneity in the relationship between `price_per_sqft` and its predictors, building on the OLS model's findings. While the OLS residuals showed no significant spatial autocorrelation, the exploratory analysis highlighted clustering in areas like Sengkang and Punggol, suggesting that the impact of features like `distance_to_mrt_meters` may vary across space. GWR's ability to model local variations makes it ideal for capturing these spatially diverse effects, with a focus on a reduced feature set to mitigate potential collinearity.

Implementation: The GWR model was fitted using `mgwr.gwr.GWR` on the full dataset (69,582 rows) after confirming no duplicate coordinates. A correlation analysis was conducted on numerical features to check for collinearity, revealing a moderate correlation between `distance_to_cbd` and `price_per_sqft_lag` (-0.6013). A reduced feature set (`distance_to_mrt_meters`, `distance_to_cbd`, `distance_to_pri_school_meters`, `price_per_sqft_lag`, `floor_area_sqm`) was

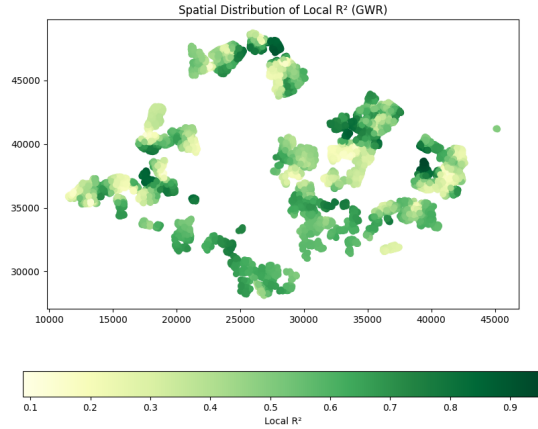


Fig. 22. Spatial Distribution of Local R^2 (GWR).

standardized and used. A fixed bandwidth of 200 nearest neighbors with a bisquare kernel was applied. Local R^2 and coefficients for `distance_to_mrt_meters` were extracted, and two visualizations were generated: a map of local R^2 and a map of local coefficients for `distance_to_mrt_meters`.

Results and Interpretation: The GWR model achieved a global R^2 of 0.7525 (adjusted R^2 : 0.7378), outperforming OLS (R^2 : 0.7076) by capturing an additional 4.5% of the variance in `price_per_sqft`. The AIC was 107,544.91, with an effective number of parameters of 540.19, reflecting the model's complexity due to local parameter estimation. The moderate correlation between `distance_to_cbd` and `price_per_sqft_lag` (-0.6013) suggests some collinearity, but the reduced feature set mitigated its impact. Local coefficients for `distance_to_mrt_meters` varied significantly, ranging from -400 to 1,000, indicating spatially diverse effects of MRT proximity on prices.

Visualization Details: Two plots were generated to evaluate GWR's spatial performance. Figure 22 shows the spatial distribution of local R^2 , ranging from 0.1 to 0.9, with a yellow-green colormap. Higher local R^2 values (0.7–0.9) are observed in central and northeastern regions, indicating better model fit in these areas. Figure 23 maps the local coefficients for `distance_to_mrt_meters` using a red-blue colormap, revealing strong negative effects (up to -400) in central areas near MRT stations and positive effects (up to 1,000) in peripheral regions, highlighting spatial heterogeneity in MRT's influence.

Conclusions: GWR significantly improves model performance over OLS, achieving a global R^2 of 0.7525 by capturing spatial heterogeneity in the relationship between `price_per_sqft` and its predictors. The spatial variation in `distance_to_mrt_meters` coefficients underscores the differing impact of MRT proximity across Singapore, with

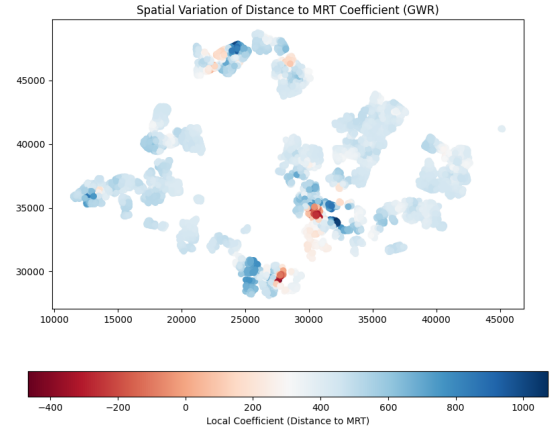


Fig. 23. Spatial Variation of Distance to MRT Coefficient (GWR).

stronger negative effects near central MRT stations. This aligns with the exploratory findings of high price densities in central areas. Future work could explore adaptive bandwidths or additional features to further enhance GWR's explanatory power.

C. Spatial Clustering and Classification

Rationale for Methods: Spatial clustering was performed to identify distinct groups of HDB flats based on their spatial coordinates and `price_per_sqft`, building on the GWR findings of spatially varying relationships. HDBSCAN was chosen over traditional clustering methods like k-means due to its ability to handle noise and detect clusters of varying densities, which is suitable for the heterogeneous spatial distribution of flats in Singapore. The analysis evaluates clusters by planning area distribution and MRT proximity to uncover patterns in price and location, complementing the earlier hotspot and regression analyses.

Implementation: HDBSCAN clustering was applied using `hdbscan.HDBSCAN` with a minimum cluster size of 50, using standardized features (`x`, `y`, `price_per_sqft`) to ensure equal weighting. The resulting clusters were added to the `GeoDataFrame`. Cluster evaluation involved computing the distribution of `planning_area_ura` within each cluster, normalized to proportions, and calculating the mean `distance_to_mrt_meters` per cluster. Three visualizations were generated: a spatial cluster map, a stacked bar chart of planning area proportions, and a boxplot of MRT proximity by cluster.

Results and Interpretation: HDBSCAN identified three clusters and 141 noise points (0.20% of the dataset). Cluster 0 (north) includes Woodlands and Yishun, with a mean `distance_to_mrt_meters` of 0.1989 (standardized), indicating farther MRT access. Cluster 1 (central-northeast) comprises Sengkang, Tampines, and Ang Mo Kio, with a mean `distance_to_mrt_meters` of -0.0799, reflecting

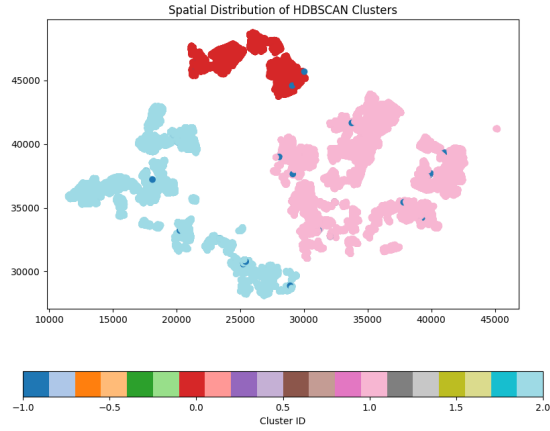


Fig. 24. Spatial Distribution of HDBSCAN Clusters.

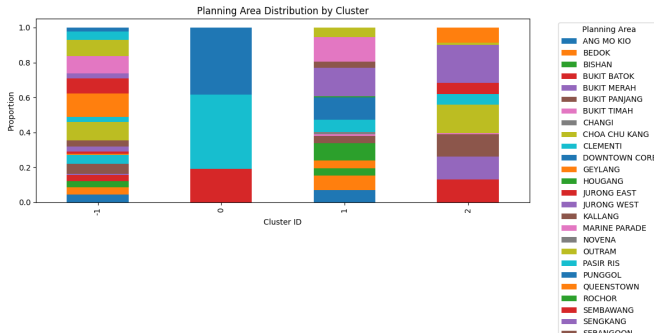


Fig. 25. Planning Area Distribution by Cluster.

closer MRT proximity. Cluster 2 (west-central) includes Bukit Batok, Bukit Merah, and Choa Chu Kang, with a mean `distance_to_mrt_meters` of 0.0266. The planning area distribution shows distinct regional groupings, with Cluster 1 aligning with high-price-density areas like Sengkang, as identified in the KDE analysis.

Visualization Details: Three plots were generated to evaluate the clusters. Figure 24 maps the spatial distribution of clusters using a categorical colormap, showing Cluster 0 in the north, Cluster 1 in the central-northeast, and Cluster 2 in the west-central regions. Figure 25 is a stacked bar chart of planning area proportions, highlighting the dominance of Sengkang in Cluster 1 (16.29%) and Woodlands in Cluster 0 (42.50%). Figure 26 presents a boxplot of `distance_to_mrt_meters` by cluster, confirming Cluster 1's closer MRT proximity and Cluster 0's greater distance, with some outliers in Cluster -1.

Conclusions: HDBSCAN clustering effectively identified three spatially distinct groups of HDB flats, reflecting regional patterns in price and MRT proximity. Cluster 1's concentration in Sengkang and Tampines, with closer MRT access, aligns with the high price densities observed in

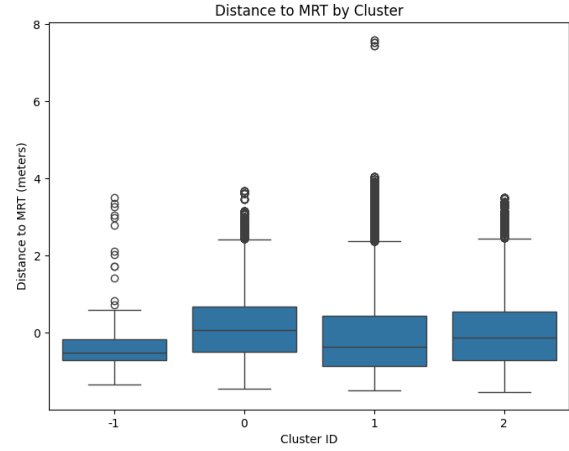


Fig. 26. Distance to MRT by Cluster.

the KDE analysis and GWR's strong local coefficients for `distance_to_mrt_meters` in these areas. The minimal noise (0.20%) indicates robust clustering. These findings can guide targeted urban planning by highlighting areas with similar price-location profiles, though future work could incorporate additional features like flat type to refine cluster interpretations.

D. Predictive Modeling with Spatial Awareness

Rationale for Methods: A Random Forest model was employed to enhance predictive accuracy for `price_per_sqft` by capturing nonlinear relationships and interactions among features, building on the linear models (OLS, GWR) and clustering insights. Random Forest's ensemble approach reduces overfitting and provides feature importance, allowing us to assess the relative impact of spatial features like `price_per_sqft_lag` and `distance_to_mrt_meters`, which were significant in GWR and clustering analyses. This method complements the spatial awareness introduced by earlier models.

Implementation: A Random Forest model was trained using `sklearn.ensemble.RandomForestRegressor` with 100 trees on the training set (55,665 samples) from the feature engineering step. The feature set included `distance_to_mrt_meters`, `distance_to_cbd`, `distance_to_pri_school_meters`, `price_per_sqft_lag`, `floor_area_sqm`, `planning_area_ura_encoded`, and `flat_type_encoded`. Predictions were made on the test set (13,917 samples), and performance was evaluated using mean absolute error (MAE), root mean squared error (RMSE), and R^2 . Feature importance was extracted, and two visualizations were generated: a bar chart of feature importance and a scatter plot of predicted versus actual `price_per_sqft`.

Results and Interpretation: The Random Forest model achieved an R^2 of 0.8681 on the test set, significantly out-

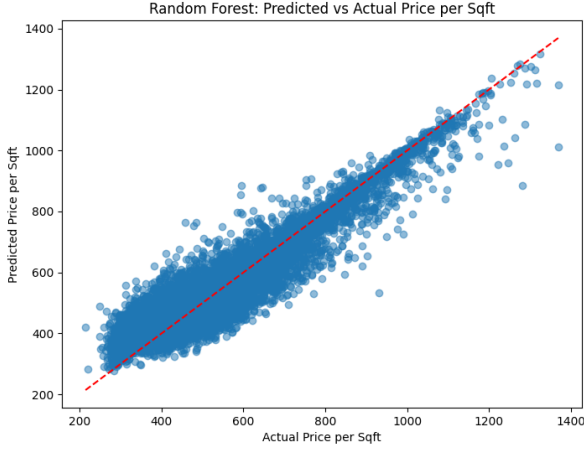


Fig. 27. Random Forest: Predicted vs Actual Price per Sqft.

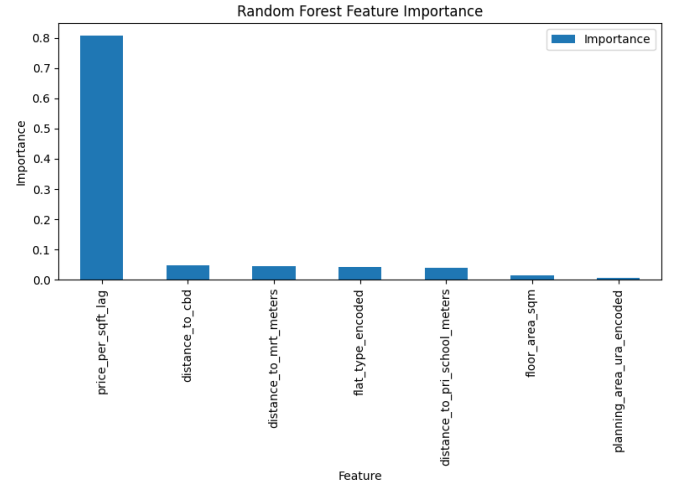


Fig. 28. Random Forest Feature Importance.

performing OLS (R^2 : 0.7076) and GWR (R^2 : 0.7525) by explaining 86.81% of the variance in `price_per_sqft`. The MAE was 39.88, and the RMSE was 55.53, indicating improved predictive accuracy over OLS (RMSE: 82.66) and reflecting the model's ability to capture nonlinear patterns. Feature importance analysis identified `price_per_sqft_lag` as the most influential feature (importance: 0.8083), followed by `distance_to_cbd` (0.0476) and `distance_to_mrt_meters` (0.0434), underscoring the critical role of spatial lag in price prediction.

Visualization Details: Two plots were generated to evaluate the Random Forest model. Figure 27 is a scatter plot of predicted versus actual `price_per_sqft`, with a red dashed line representing perfect prediction. The points closely follow the line, confirming high predictive accuracy, though some dispersion is observed at higher price ranges. Figure 28 is a bar chart of feature importance, highlighting the dominance of `price_per_sqft_lag` (0.8083) over other features, with spatial proximity features like `distance_to_cbd` and `distance_to_mrt_meters` also contributing significantly.

Conclusions: The Random Forest model substantially improves predictive performance (R^2 : 0.8681) over OLS and GWR, demonstrating the value of capturing nonlinear relationships in `price_per_sqft` prediction. The dominance of `price_per_sqft_lag` in feature importance aligns with the clustering findings, where neighborhoods like Sengkang (Cluster 1) showed high price densities and close MRT proximity. These results suggest that spatial lag effects are critical for accurate price prediction in Singapore's HDB market. Future work could explore hyperparameter tuning or ensemble methods to further enhance performance.

E. Conclusion and Final Remarks

The spatial machine learning analysis provided a comprehensive understanding of the factors driving `price_per_sqft` in Singapore's HDB

resale market, with a focus on spatial dynamics. Feature engineering effectively incorporated spatial predictors, such as `distance_to_mrt_meters` and `price_per_sqft_lag`, setting the foundation for modeling. The OLS regression model (R^2 : 0.7076) established a baseline, revealing the importance of spatial features but also the limitations of assuming spatially constant relationships. Geographically Weighted Regression (GWR) improved performance (R^2 : 0.7525) by capturing spatial heterogeneity, with local coefficients for `distance_to_mrt_meters` highlighting stronger price effects near central MRT stations. Spatial clustering via HDBSCAN identified three distinct groups, aligning high-price areas like Sengkang (Cluster 1) with close MRT proximity, reinforcing the exploratory findings of price clustering. Finally, the Random Forest model achieved the highest predictive accuracy (R^2 : 0.8681), with `price_per_sqft_lag` as the dominant feature (importance: 0.8083), underscoring the critical role of neighborhood effects in price prediction.

These findings confirm that spatial factors, particularly proximity to CBD and MRT and neighborhood price trends, are key drivers of HDB resale prices, with effects varying across Singapore. The progression from linear models to spatially-aware regression and nonlinear ensemble methods highlights the complexity of spatial relationships in urban housing markets. Future work could explore adaptive bandwidths in GWR, incorporate additional features like commercial hub proximity in clustering, or leverage ensemble methods with spatial constraints to further enhance predictive performance. This analysis provides actionable insights for urban planners and policymakers, emphasizing the need for spatially targeted strategies to address housing price disparities in Singapore.

IV. CHALLENGES

This exploratory spatial analysis encountered several issues, multiple listings per flat skewed spatial patterns, so we filtered to the latest transaction using identifier columns; verifying

x, y alignment with SVY21 (EPSG:3414) was complex but resolved through careful checks; KDE bandwidth selection risked overfitting or oversmoothing, addressed by testing multiple values; the GWR model failed with a "singular matrix" error due to similar data points, fixed by increasing bandwidth and tweaking coordinates; balancing features in GWR required standardizing data and capping outliers; HDBSCAN clustering needed tuning, settled on an appropriate cluster size to align with planning areas and MRT locations; and computational intensity of KDE, LISA, and visualizations caused long processing times.

V. AUTHORS' CONTRIBUTIONS

A. Adithya Nangarath

Author 1 was responsible for the data preprocessing and exploratory statistical analysis components of this report. This included cleaning and preparing the HDB resale transaction dataset, ensuring a unique spatial snapshot by filtering to the latest listing per flat, and conducting spatial statistical analyses such as Moran's I, LISA, variograms, KDE, and Temporal Hotspot Analysis to uncover clustering patterns, spatial ranges of influence, and price density gradients.

B. Kalahasti Venkata Nikhilesh

Author 2 led the spatial machine learning segment, encompassing feature engineering (spatial lags, distance features), OLS regression, Geographically Weighted Regression (GWR), HDBSCAN clustering, Random Forest modeling, and the synthesis of findings in the conclusions. This work built on the exploratory analysis to model and predict `price_per_sqft`, highlighting the role of spatial factors like MRT proximity and neighborhood effects.

Both authors collaborated to ensure the integration of statistical and machine learning insights, providing a cohesive analysis of Singapore's HDB resale market.

VI. ACKNOWLEDGMENTS

We would like to express our heartfelt gratitude to Professor Jaya Sreevalsan Nair and Professor Ramesh Kestur for their exceptional guidance and teachings in the SpatioTemporal Data Analytics course at IIITB. We are deeply grateful for the opportunity to undertake this assignment under their mentorship. We also acknowledge the use of ChatGPT and Grok 3 for obtaining information and generating segments of text and code; however, all analysis, interpretations, and intellectual contributions were solely our own, driven by the knowledge and skills acquired during the course.

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